Lesson 5: Experimental Methodology

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell’Aquila

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Before we start

- Exercise 4, which we discuss now
- The threshold you chose, and how you chose it

We broke a promise on purpose

- Lesson 4 chose a threshold **by looking at the test set**
- Lesson 1 forbade exactly that
- Today we pay the debt

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Today: how to tell whether it works

- One split is a lottery, and how wide the lottery is
- Cross-validation: what it estimates, and what it does not
- Bias, variance, and the floor you cannot go below
- Learning curves: which fix to buy
- Leakage that survives cross-validation

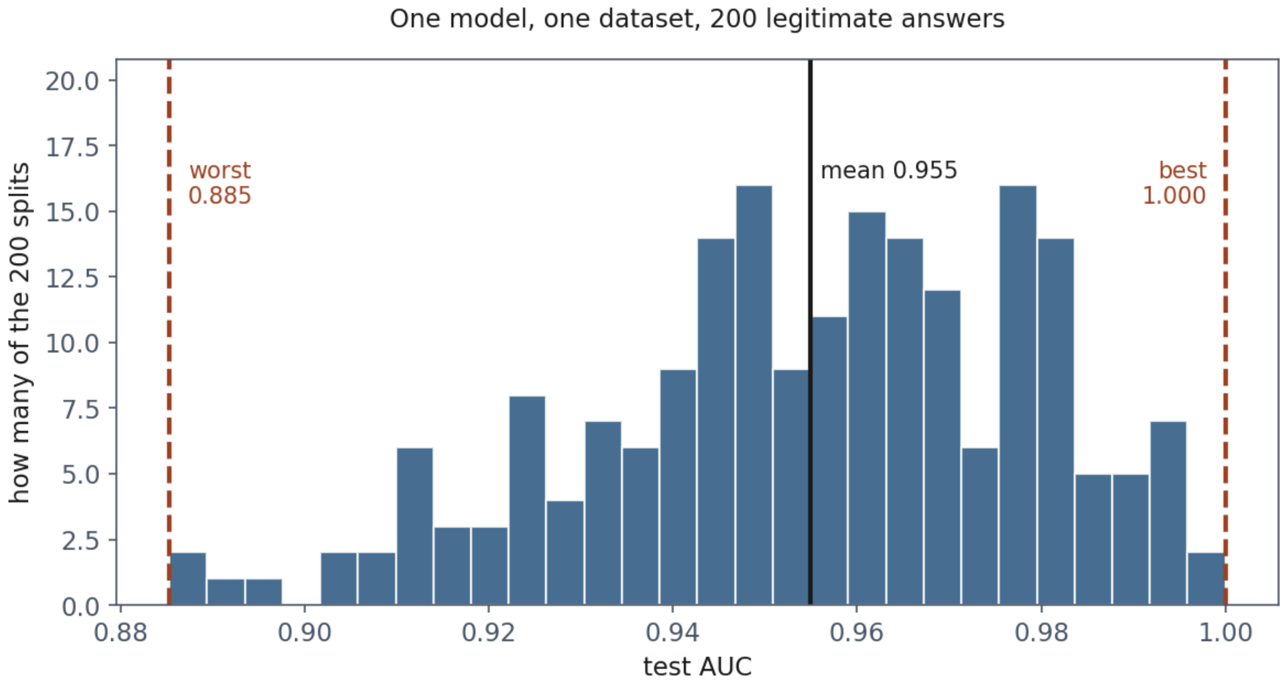
A test score is a measurement

- You would not report a poll of forty people to three decimals
- Holding data out makes the estimate **unbiased**
- It does nothing to make it **precise**

The fleet, cut down

- Lesson 4’s drives, reduced to **800**, of which **29 fail**
- Not to exaggerate: 800 rows with 29 positives is the ordinary case

200 honest splits, one dataset



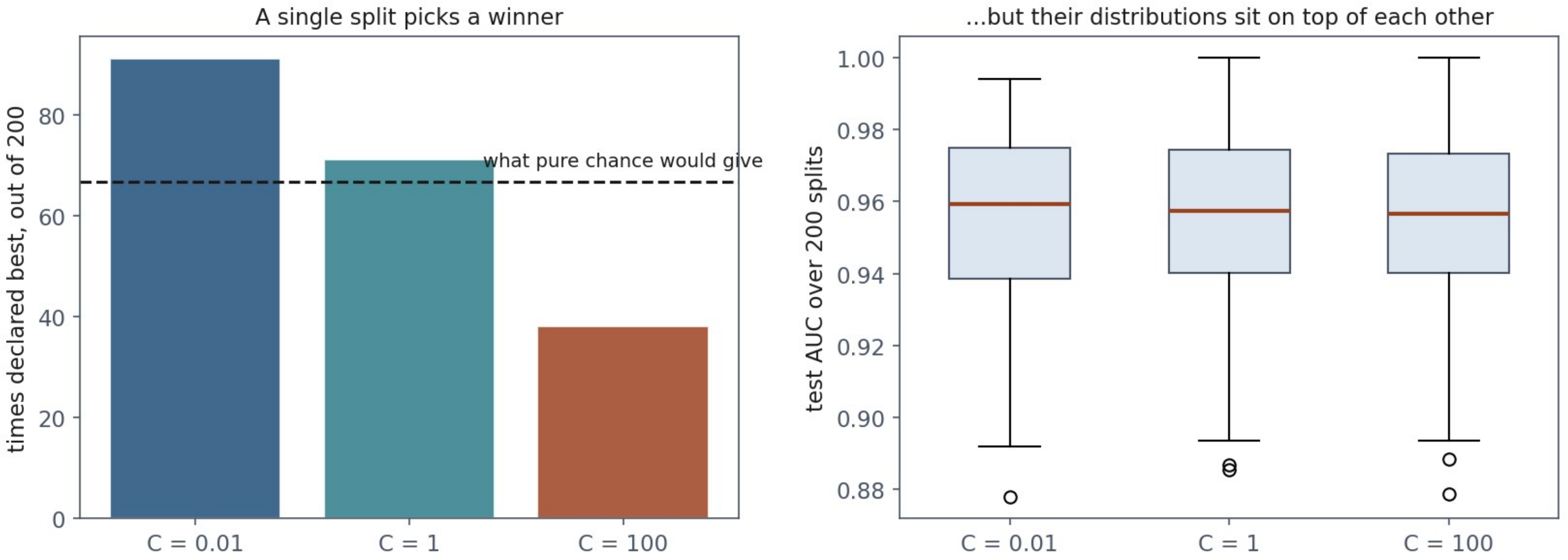
Why so wide?

- Each test set holds **7 failures**
- Every AUC (area under the receiver operating characteristic curve) rests on seven

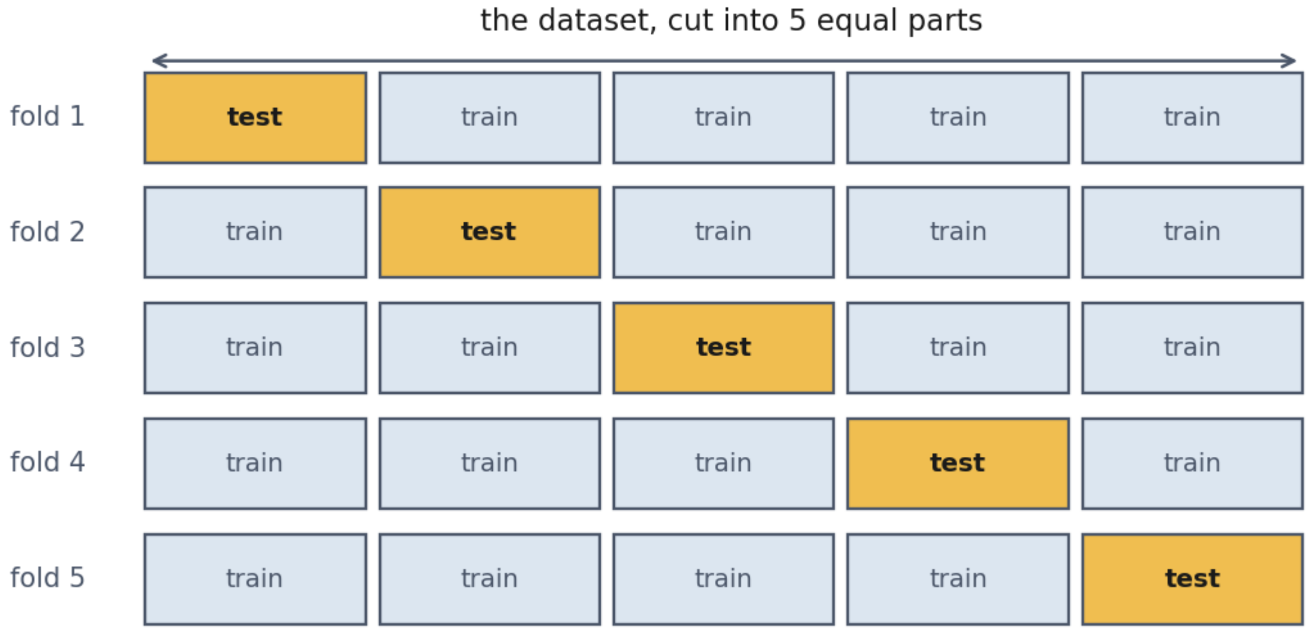
Now the expensive part: choosing

Model	Mean AUC	Times declared best
C = 0.01	0.9549	91
C = 1	0.9549	71
C = 100	0.9544	38

The same three models, drawn honestly



k-fold cross-validation



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What it actually estimates

- The performance of a **procedure**, not of one fitted model
- Slightly **pessimistic**: each fold trains on 80% of the data

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Report the spread, but not as a confidence interval

- Any two training folds share **three quarters** of their rows
- The scores are correlated, so the usual standard error is optimistic

Stratify when a class is rare

Fold	Plain KFold	StratifiedKFold
1	7 failures of 160	5 of 160
2	**1 failure of 160**	6 of 160
3	7 of 160	6 of 160
4	5 of 160	6 of 160
5	9 of 160	6 of 160

How much does it help?

	Single split	5-fold CV
worst	0.9119	0.9439
best	1.0000	0.9588
spread	0.0881	0.0149

How many folds?

- **k = 5 or 10** in practice
- Larger k: less pessimistic, more expensive, folds more correlated
- k = m is **leave-one-out**

When the rows are not independent

- Ten readings from the **same drive**, split at random
- Nine in training, one in test
- The model recognises the drive, not the failure

Split by the thing you must generalise to

- Same entity in two folds → **GroupKFold**
- Predicting the future → **TimeSeriesSplit**

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Notebook 1, live

- The split lottery, the three identical models, k-fold by hand

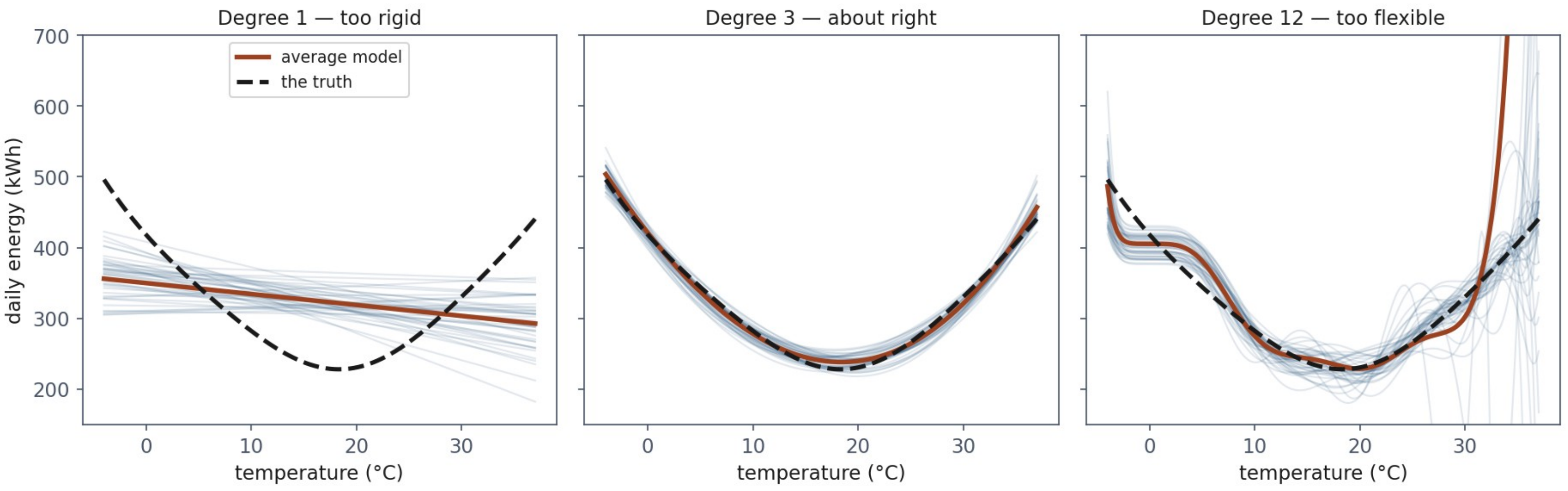
Break

- Ten minutes

Where does the error come from?

- **Bias**: the average model is wrong
- **Variance**: the individual models disagree
- **Noise**: the target itself is uncertain

Three hundred parallel universes



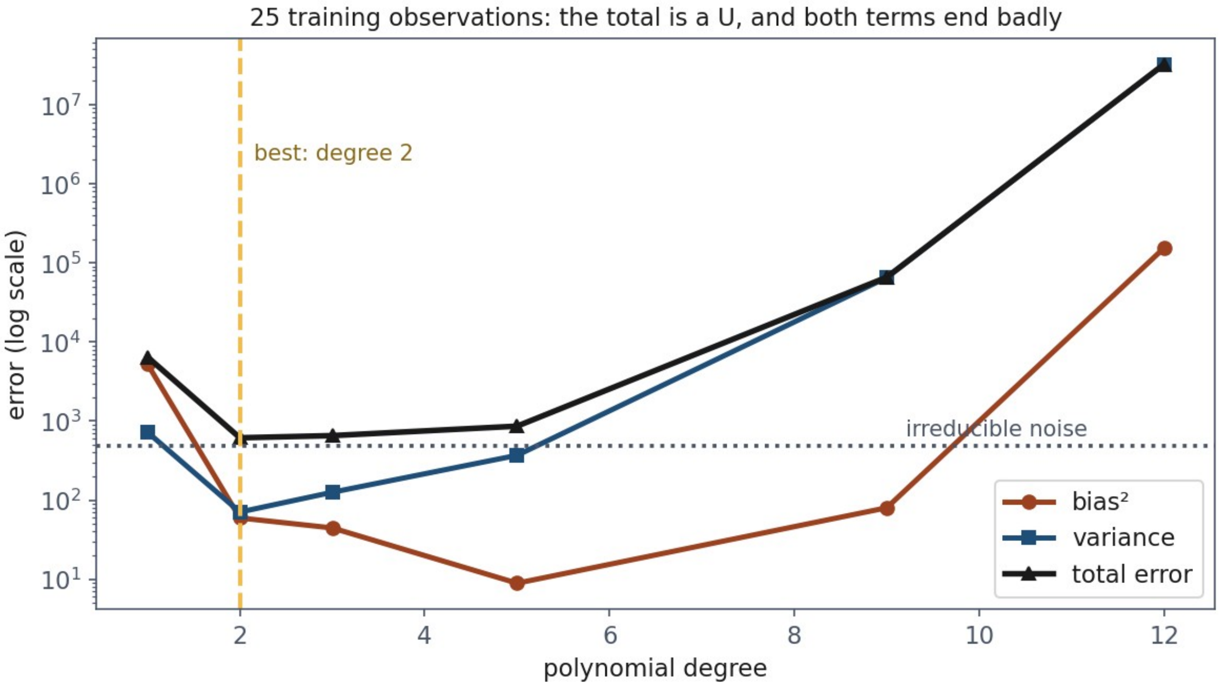
The decomposition

$$\mathrm{E}\left[(y - \hat{f})^2\right] = \mathrm{bias}^2 + \mathrm{variance} + \sigma^2$$

Measured, on 25 observations

Degree	$bias^2$	Variance	Noise	Total
1	5,250.8	723.8	484.0	6,458.6
2	59.6	70.8	484.0	**614.3**
5	8.9	368.9	484.0	861.8
12	153,537.3	32,174,344.9	484.0	32,328,366.2

The picture everyone has seen, now measured



The floor you cannot go below

- Noise variance is **484** at every complexity
- A reported error below the floor means **contamination**, not excellence

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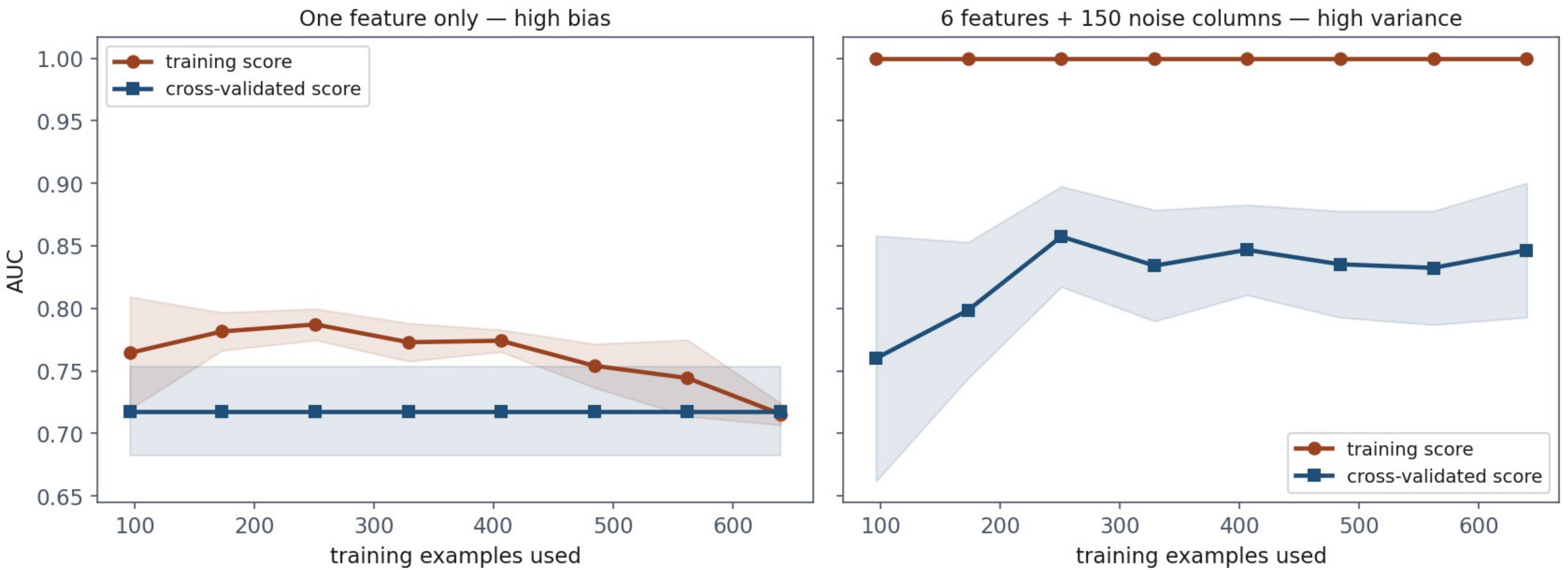
The best model depends on how much data you have

Degree	m = 25	m = 60	m = 150
2	**614.3**	570.6	555.3
5	861.8	**540.9**	**506.9**
12	32,328,366.2	7,626.9	899.0

Learning curves: the same diagnosis, without knowing the truth



Which fix to buy

- Curves **meet low** → bias. More data will **not** help
- Wide gap, validation **still rising** → variance. More data **will**

Notebook 2, live

- Measuring bias and variance; the learning curves

Now: what cross-validation does not protect you from

- 2,000 columns of **pure random noise**
- No relationship to the label whatsoever
- The honest score of any model on it is **0.500**

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The code almost everyone writes

```
selector = SelectKBest(f_classif,
k=10).fit(noise, y)
chosen = noise.loc[:,
selector.get_support()]
cross_val_score(model, chosen, y,
cv=folds)
```

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0.931

- Cross-validated AUC **0.931** on data with no signal
- Four of five folds between 0.93 and 0.98

Fix it, and look what happens

- Selection moved **inside** the pipeline, refitted per fold
- 20 different fold seeds: **0.658**
- The truth is still 0.500

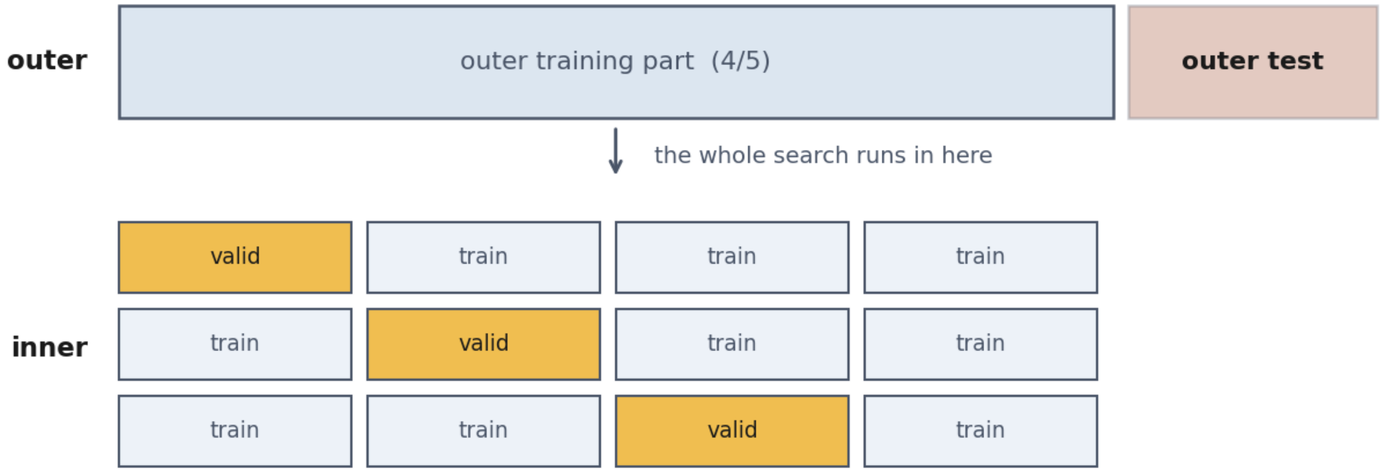
Why it does not return to 0.5

- With 2,000 columns and 800 rows, some columns match the label **by accident**
- That accident belongs to the **sample**, not to the split
- Every fold contains it

Hyperparameter search is the same problem

	AUC
best of 25 candidates	**0.7999**
average candidate	0.7265
nested cross-validation	0.6699
the truth	0.500

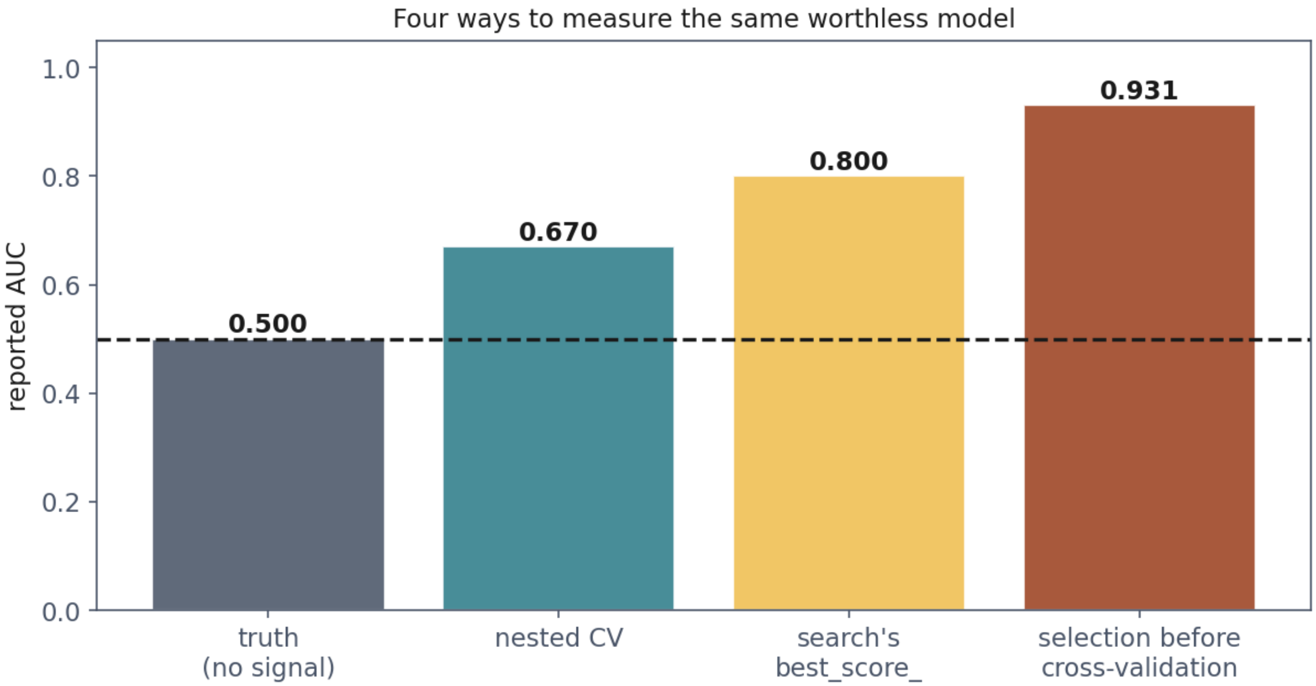
Nested cross-validation



The inner loop may overfit itself as much as it likes.

The outer test block was never part of it: that is the number you report.

Four ways to measure the same worthless model



The debt, paid

Policy	Cost on the test set
threshold 0.50, the default	10,540 EUR
threshold chosen on **validation**	**4,700 EUR**
threshold chosen on **test**	3,300 EUR

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Three sets, three jobs

- **Train**: fit the parameters
- **Validation**: make every choice: model, hyperparameters, threshold
- **Test**: report. Touched **once**

A fixed seed buys repeatability, not stability

- 30 seeds, all perfectly reproducible: **0.913 to 1.000**
- `AUC 1.000 (random_state=3)` is reproducible **and** misleading

What to fix, and what to write down

- Every `random_state`, and `np.random.default_rng(seed)`
- **Library versions**: defaults change between releases
- What you could **not** fix: threads, GPU non-determinism

Notebook 3, live

- The 0.931, the fix, the nested search

What to put in a report

- The metric, the **spread**, and how it was estimated
- How many **positives** the test set held
- Every seed, and every library version
- Which choices were made on which data

What to take away

- **A test score is a measurement.** 0.885 to a perfect **1.000**, same model
- **Choosing on one split chooses noise**
- **bias^2 + variance + noise** is an identity, and the noise floor never moves
- **Curves that meet low mean bias**: more data will not help
- **Cross-validation protects against fold leakage, nothing more**

Homework

- **Exercise 5**, discussed at the start of **Friday 30 October**
- A result that is too good. Find out why, then produce the honest number